

**UNIVERSITY OF THE PUNJAB, LAHORE**

Department of Computer Science



## **Hospital Management System**

### **A Software Engineering Project Report**

**Prepared by: Sara Shabir**

**Project Supervisor: Dr.Nabeel Tahir**

# HOSPITAL MANAGEMENT SYSTEM

## Introduction:

### Purpose:

Efficient management of the clinical, financial, and administrative aspects of a healthcare facility is the aim of a hospital management system (HMS). HMS seeks to improve patient care, simplify procedures, and maximize resource use. It functions as a centralised platform for the management of appointments, patient data, inventory, billing, and provider-to-provider communication. Through the automation of repetitive operations and provision of decision support tools, HMS enhances workflow effectiveness, minimizes errors, and guarantees regulatory compliance. It also encourages patient participation, backs data-driven decisionmaking, and makes it easier for healthcare delivery to continuously improve quality. All things considered, HMS is essential to improving the general efficacy, efficiency, and calibre of hospital healthcare services.

### Scope of the Project:

A Hospital Management System (HMS) project's scope includes, but is not limited to, developing software modules for electronic health records (EHR), patient management, appointment scheduling, billing and invoicing, inventory management, and reporting. It also involves integrating these modules into an organised structure that satisfies the particular demands and specifications of the medical facility. In addition, user training, data migration from current systems, customisation to fit certain workflows, and continuous technical support and maintenance are all included in the project scope.

## References:

### Books:

- *Software Requirements (Microsoft) Second Edition* By **Karl E. Wiegers**
- *Fundamentals of Database System* By **Elmasri**
- *Software Requirements and Specifications: A Lexicon of Practice, Principles and Prejudices (ACM Press)* by **Michael Jackson**
- *Software Engineering: A Practitioner's Approach Fifth Edition* By **Roger S. Pressman**

### User Roles:

## 1. Generic Authentication Module (Login & Logout)

This is a centralized, shared gateway used by every single person working in the hospital. Instead of building separate login screens for each department, everyone enters through this secure gate.

### **Generic Login Features:**

System Inputs: Every employee enters their unique Username/Email and Password.

Role-Based Access Control (RBAC): Upon hitting submit, the system cross-references the credentials with the database to check the user's role (such as Role = 'Doctor', Role = 'Receptionist').

Dynamic Redirection: Instead of landing on a generic homepage, the system dynamically reroutes the user to their specific dashboard based on their role. A nurse sees clinical queues, while an accountant sees billing panels.

Session Initialization: The system generates a secure session token (like a JWT) that tracks which station the user is logged into and logs the exact entry timestamp for auditing.

### **Generic Logout Feature**

Session Destruction: Clicking Logout instantly destroys the active session token, preventing anyone else from tampering with patient records at that computer station.

**Idle Auto-Logout (Security Guard):** If a machine is left untouched for more than 10 minutes (e.g., a doctor leaves their desk for an emergency), the system automatically triggers a logout to protect sensitive healthcare data.

When a patient walks into the hospital, their data ripples through the system from one user role to another in a continuous, linked chain.

## **Role 1: The Receptionist (The Data Initiator)**

The first point of physical contact. This role initializes the data flow.

Primary Feature: Patient Registration & Encounter Check-In.

When a patient arrives, the Receptionist captures their core bio-data (Name, Age, Contact) or searches their existing ID. The moment the Receptionist clicks Generate Token / Check-In, this patient's profile is automatically pushed directly onto the Nurse's Triage Queue in real-time.

## **Role 2: The Nurse (The Vital Recorder)**

The clinical buffer between registration and consultation.

Primary Feature: Vitals Logging (Blood Pressure, Pulse, Temperature, Weight).

The Nurse selects the patient from the queue generated by the Receptionist. The Nurse records the vital stats. The moment the Nurse clicks Save Vitals, this data attaches itself to the patient's current session and instantly populates the Doctor's Active Consultation Queue.

## **Role 3: The Doctor (The Core Evaluator)**

The central decision-maker who analyzes previous data and directs future actions.

Primary Feature: Patient Master Record View & Computerized Order Entry.

When the Doctor opens the patient's record, they instantly see the Receptionist's demographic info and the Nurse's freshly recorded vitals on one screen. The Doctor's actions then branch into two critical links:

If the Doctor clicks Order Blood Test, that specific test request immediately appears as an active task on the Lab Technician's Worklist.

When the Doctor types out an electronic prescription and signs it, the medication details are beamed straight to the Pharmacist's Dispensing Queue.

## **Role 4: The Lab Technician (The Diagnostic Provider)**

The investigator behind the scenes.

Primary Feature: Test Result Processing & Specimen Logging.

The Link to Next Role: The Lab Technician doesn't need paper forms; they see the order pre-filled by the Doctor. Once the sample is tested, the Tech inputs the numerical results or uploads digital images. Saving this report automatically sends a high-priority alert directly to the Doctor's Dashboard, allowing the Doctor to re-evaluate the patient's record with updated data.

## **Role 5: The Pharmacist (The Medicine Dispenser)**

The gatekeeper of hospital inventory.

Primary Feature: Prescription Verification & Stock Management.

When the patient walks up to the pharmacy counter, the Pharmacist simply pulls up the patient's ID. The exact prescription written by the Doctor appears on screen. As the Pharmacist clicks Dispense, two backend events happen simultaneously: the hospital's central inventory auto-deducts the stock, and the cost of those specific medicines is pushed instantly to the Billing Accountant's Pending Charges.

## **Role 6: The Billing Accountant (The Final Closer)**

The operational finale of the patient's visit.

Primary Feature: Automated Charge Capture & Invoice Generation.

The Link to All Roles: The Accountant does not manually type out what services the patient received. Instead, the HMS has tracked the entire journey. The Accountant opens the billing file, and the system pulls a compiled list of financial items: the Receptionist's registration fee + the Nurse's triage fee + the Doctor's consultation fee + the Lab Tech's test costs + the Pharmacist's medicine totals. The Accountant processes the unified invoice, accepts payment, and marks the patient status as Discharged.

## **The Data Flow Summary**

Login (Universal) → Receptionist (Creates Profile) → Nurse (Appends Vitals) → Doctor (Views Record & Orders Services) → Lab Tech (Uploads Reports) → Pharmacist (Dispenses Stock) → Accountant (Collects Consolidated Bill) → Logout (Secures Station)

# Features of Hospital Management System

## 🔒 1. The Generic Authentication Module (Universal Gate)

Before anyone can perform their duties, they must pass through this core identity system. Every System User should be able to:

- Log In securely using their unique email/username and encrypted password.
- Be redirected dynamically to their designated department dashboard based on their database role identifier (Doctor, Nurse, Patient, etc.).
- Log Out manually at any moment to immediately wipe active session data from the browser/terminal.
- Be Logged Out automatically by the system after 10 minutes of complete inactivity to prevent unauthorized data access at unattended computers.

## The Receptionist (Front Desk)

The front desk is the system's absolute entry point. They initialize the data chain.

- Should be able to search the database using a national identity card number (CNIC), phone number, or unique Patient ID to see if an arriving patient already exists in the system.
- Should be able to create a new patient profile (inputting Name, Age, Gender, Emergency Contact) if the patient is visiting for the first time.
- Should be able to view real-time doctor availability calendars to book, reschedule, or cancel consultation slots.
- Should be able to check-in a patient upon arrival, automatically generating an electronic token number that immediately inserts that patient into the Nurse's Waiting Queue.

## The Nurse (Triage & Ward Management)

The bridge between registration and physical check-ups.

- Should be able to view the active waiting queue of checked-in patients sent over by the Receptionist.

- Should be able to select a patient from that queue and open their specific encounter file to record vital signs (Blood Pressure, Heart Rate, Temperature, Respiratory Rate, and Weight).
- Should be able to flag high-risk vitals (e.g., critically high blood pressure), which visually highlights that patient in red on the doctor's screen.
- Should be able to save these vitals, which automatically removes the patient from the Nurse's queue and instantly pushes them into the Doctor's Active Consultation Dashboard.
- Should be able to view and log medicine administration schedules for admitted patients in wards, ticking off doses as "Administered" based on direct doctor orders.

### **The Doctor (Clinical Decision Maker)**

The central core of the clinical ecosystem.

- Should be able to view a live, dynamically updating queue of patients who have completed triage with the Nurse.
- Should be able to open and view the complete historical Patient Record, including demographics, past diagnoses, chronic conditions, known allergies, and previous medication histories.
- Should be able to view the fresh vitals recorded by the Nurse at the top of the screen before the patient even speaks.
- Should be able to write and save digital clinical case notes detailing current symptoms and physical exam findings.
- Should be able to order laboratory tests or radiology imaging with a single click, which instantly beams a digital request to the Lab Technician's active worklist.
- Should be able to search and prescribe medications using an integrated drug database. The system should alert the doctor if a drug conflicts with the patient's logged allergies or if the Pharmacist's live inventory shows it is completely out of stock.
- Should be able to close the session, which auto-locks the clinical notes for security and

sends the digital files forward to the Pharmacist and Billing Accountant. **The Lab**

**Technician / Radiologist (Diagnostics)** The technical investigative team.

- Should be able to view an automated, incoming list of diagnostic test requests submitted by various Doctors, sorted by urgency (Routine vs. STAT/Emergency).
- Should be able to log sample collection details (e.g., "Blood sample collected at 10:15 AM").

- Should be able to enter quantitative numerical data or type up qualitative text analysis for test results.
- Should be able to upload digital attachments, such as PDF reports, X-ray files, or MRI images directly into the system.
- Should be able to hit 'Publish Results', which locks the record, instantly sends a

notification back to the ordering Doctor's dashboard, and updates the Patient's portal. **The**

### **Pharmacist (Dispensing & Stock Control)**

The controller of hospital medical inventory.

- Should be able to view an active queue of electronic prescriptions issued by Doctors, searchable by Patient ID or token number.
- Should be able to view the exact dosage, quantity, and specific instructions written by the doctor (eliminating paper-reading errors).
- Should be able to click 'Dispense Medication', which automatically triggers two background events: it subtracts the exact quantity from the central pharmacy stock ledger, and it sends the cost items directly to the Billing Accountant's dashboard.
- Should be able to manage inventory, including adding new batches of medicines, tracking manufacturing/expiry dates, and setting up automatic low-stock alerts.

### **The Billing Accountant (Finance & Checkout)**

The final stage of the patient's data journey.

- Should be able to open a unified invoice screen for any checking-out patient.
- Should be able to view an automatically compiled, itemized breakdown of costs pulled dynamically from every previous role:
  - *From Receptionist:* Registration/Token Fee.
  - *From Nurse:* Consumables used during triage.
  - *From Doctor:* Consultation and check-up fees.
  - *From Lab Tech:* Costs of all completed blood tests/X-rays.
  - *From Pharmacist:* Total cost of dispensed medications.
- Should be able to apply insurance policies or pre-approved hospital discounts to the total amount.
- Should be able to process payments (Cash, Card, Insurance Claim) and change the patient status in the database to Discharged, officially closing the visit loop.

### **The Patient (The Portal User)**

- The consumer role with strictly restricted, read-only permissions.
- Should be able to view and download their digital laboratory and radiology reports the exact moment they are finalized by the Lab Technician.
- Should be able to view their active and past electronic prescriptions, allowing them to recall proper medication dosages at home.
- Should be able to view their past billing receipts and current outstanding balances.
- Should be able to request or book future appointments online, which directly links back into the Receptionist's master scheduling calendar.
- Should NEVER be able to modify, delete, or add any data within medical, financial, or staff records. Their access is strictly Read-Only for their own personal data.

### **The System Administrator (IT & Security)**

The invisible superuser who manages the software, not the hospital operations.

- Should be able to create, modify, or deactivate user accounts for staff members (e.g., creating a login for a newly hired doctor).
- Should be able to assign or revoke roles and permissions (Role-Based Access Control), ensuring a nurse cannot access billing configurations and an accountant cannot edit a doctor's clinical notes.

## **4. System Requirements Specifications (SRS)**

Below is the design of our student registration system that included features wise use case, activity, and sequence diagram. Finally, we included system level class diagram as well as physical schema of the database.

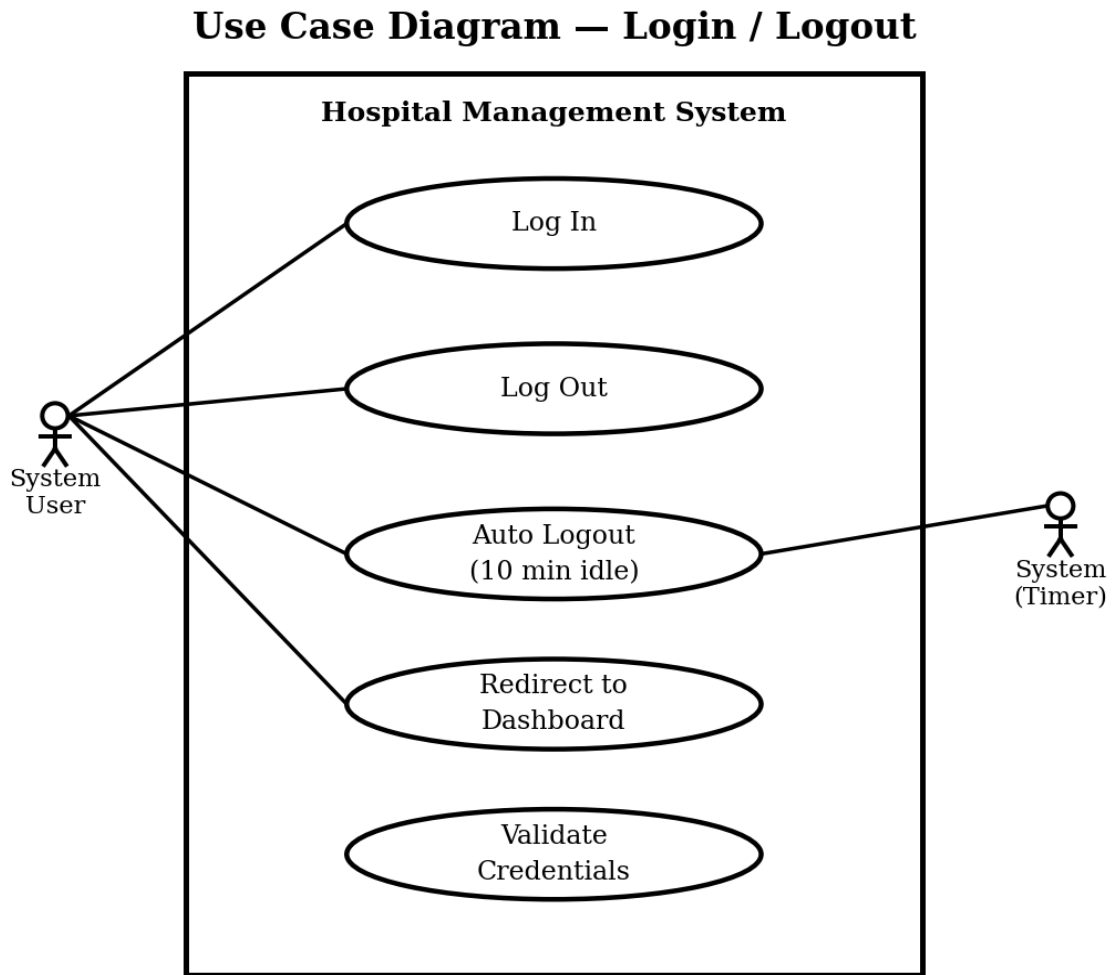
4.1 Every System User should be able to Log In securely using their unique email/username and encrypted password.

## **Hospital Management System**

## Feature 1: Login / Logout (Authentication Module)

### 1.1 Use Case Diagram

System User can Log In, Log Out, and is automatically logged out after 10 minutes of inactivity by the System Timer.

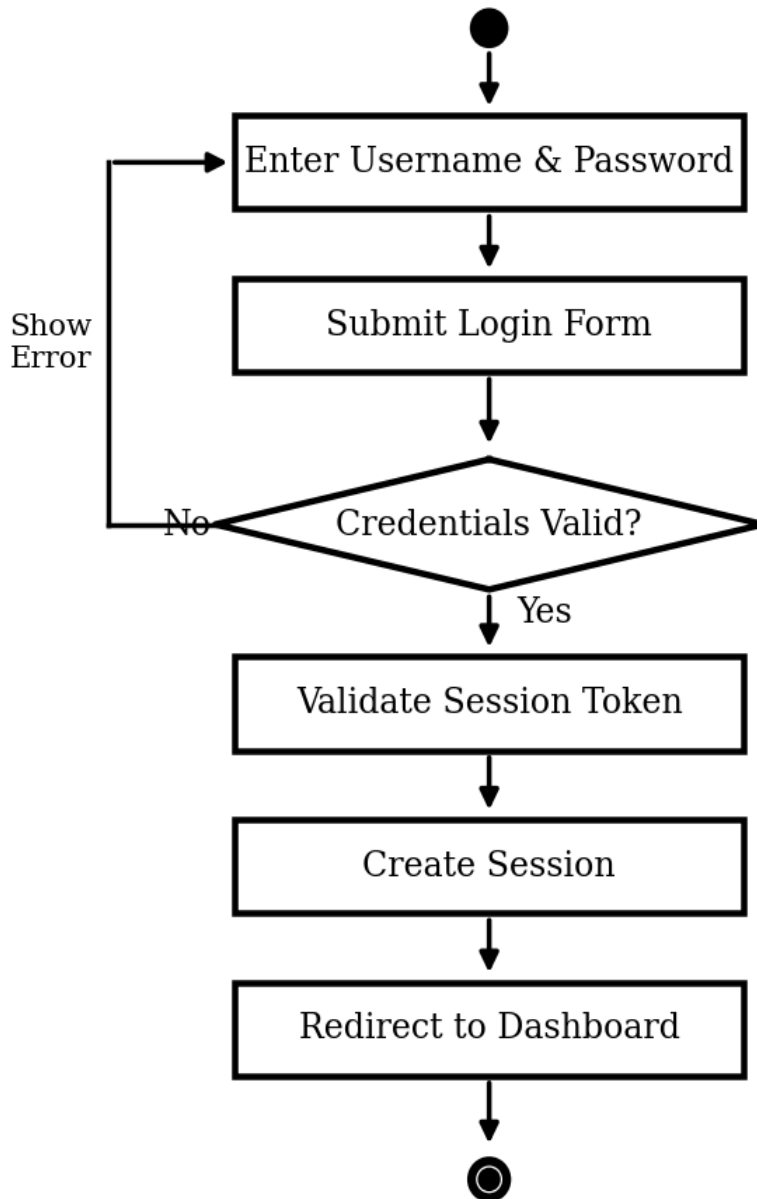


p

### 1.2 Activity Diagram — Login

Invalid credentials show an error and loop back. Valid credentials create a session and redirect to the dashboard.

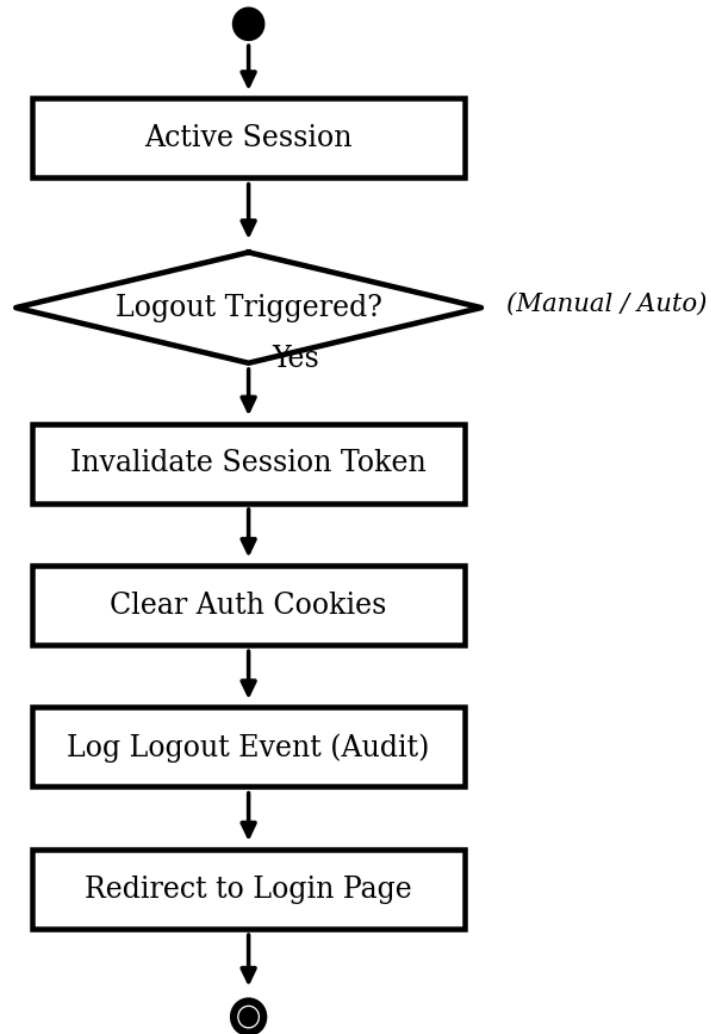
## Activity Diagram — Login



### 1.3 Activity Diagram — Logout

Both manual logout and automatic logout after 10 minutes. Session is invalidated, cookies cleared, event logged, user redirected.

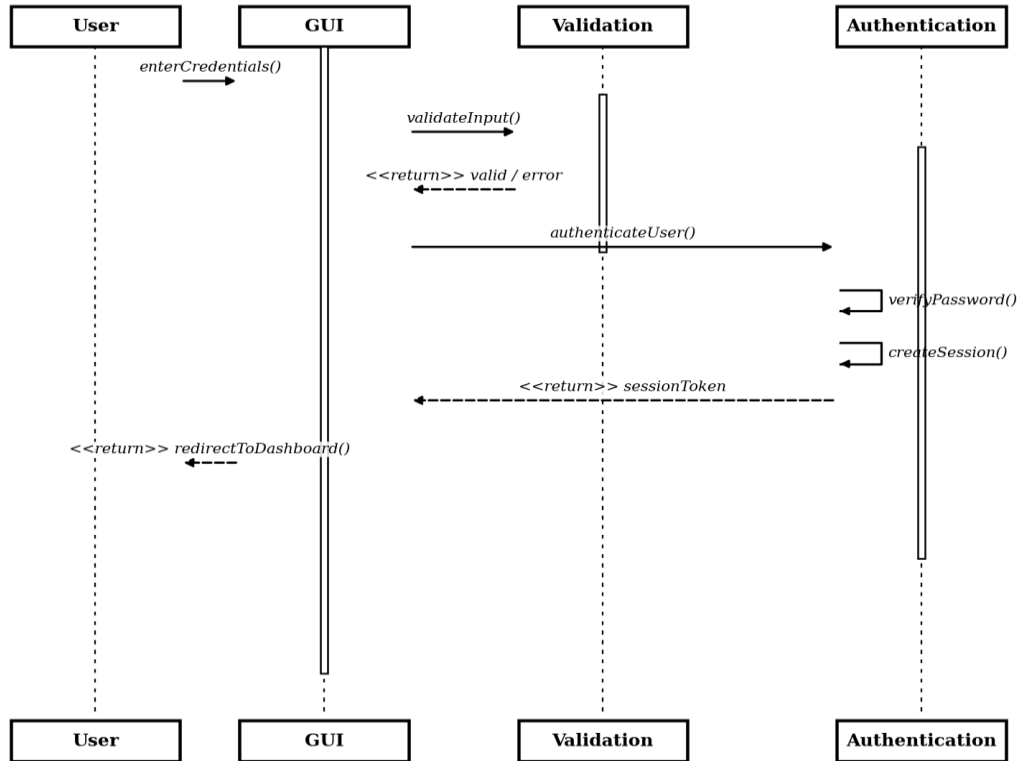
## Activity Diagram — Logout



### 1.4 Sequence Diagram — Login

Message flow: User → GUI → Validation → Authentication. Method calls with () notation and dashed return arrows.

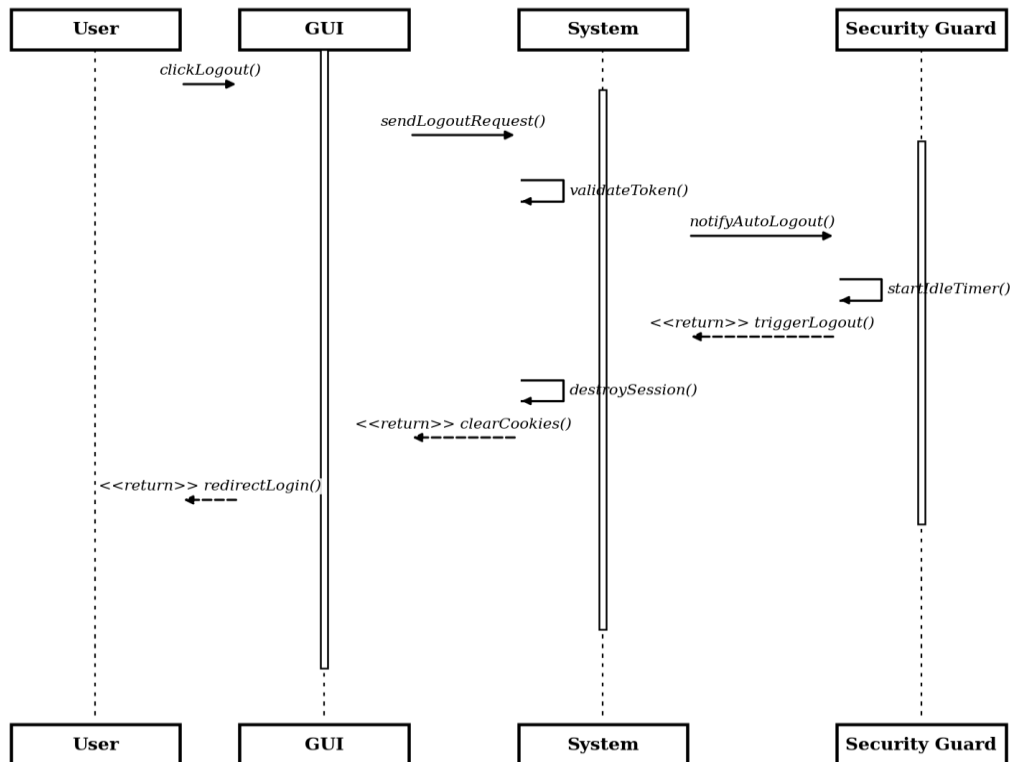
### Sequence Diagram — Login



### 1.5 Sequence Diagram — Logout

User → GUI → System → Security Guard. Security Guard manages idle timer and triggers auto-logout.

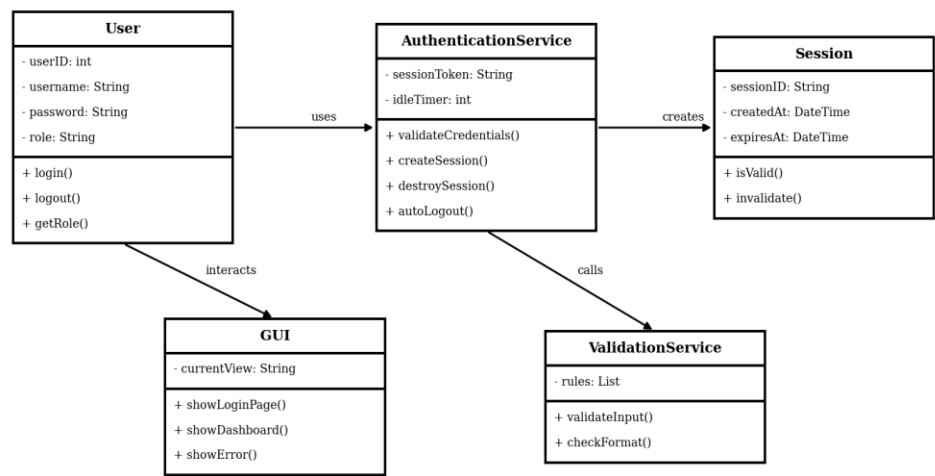
### Sequence Diagram — Logout



1.6 Class Diagram

User, AuthenticationService, Session, GUI, and ValidationService classes with their attributes and methods.

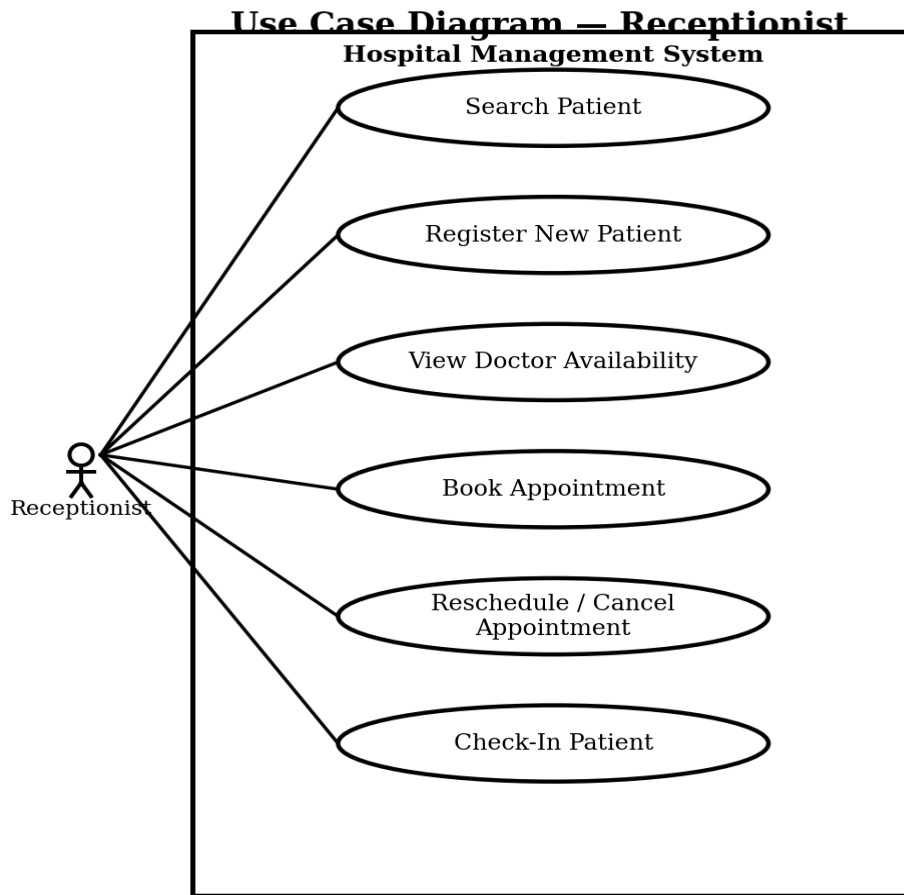
Class Diagram – Login / Logout



Feature 2: Receptionist (Front Desk)

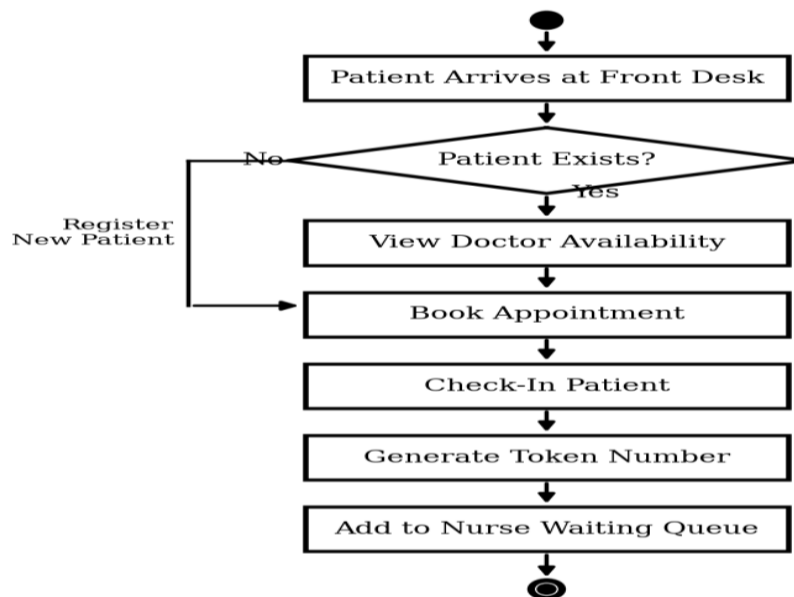
2.1 Use Case Diagram

Receptionist can search patients, register new patients, view doctor availability, book appointments, check in patients, and generate token numbers.



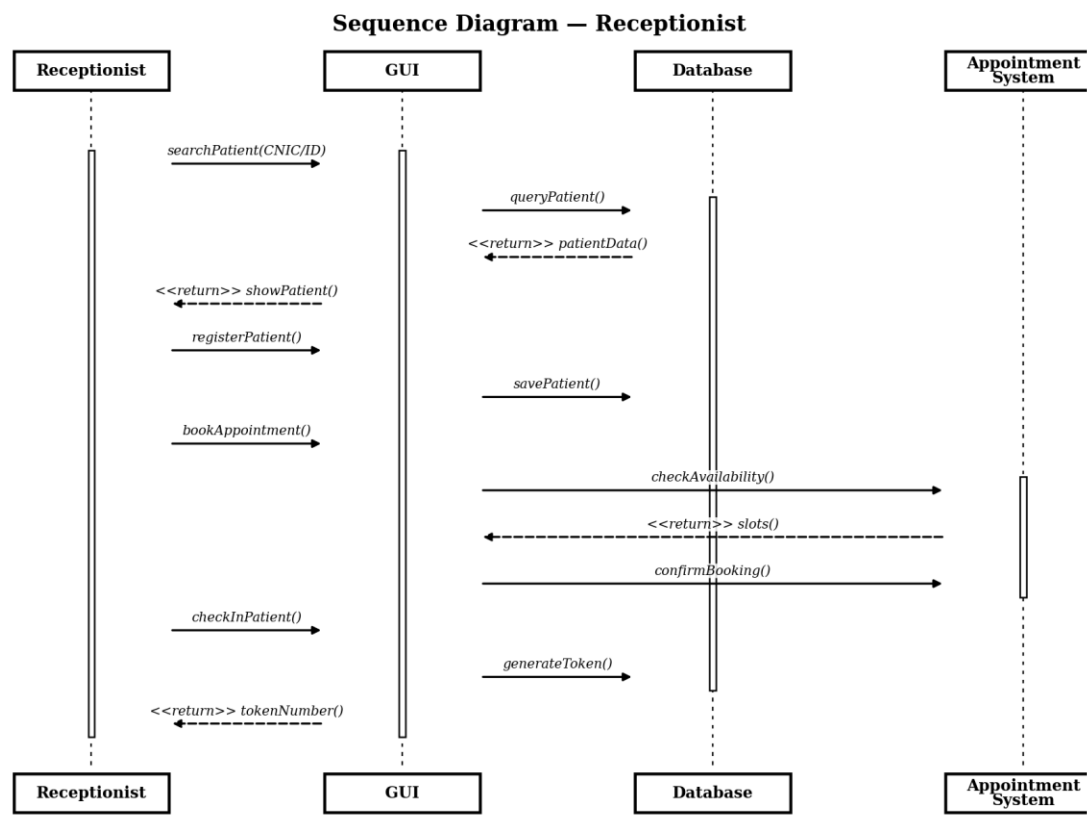
## 2.2 Activity Diagram

Patient arrives → search database → register if new → book appointment → check in → generate token → add to Nurse queue.



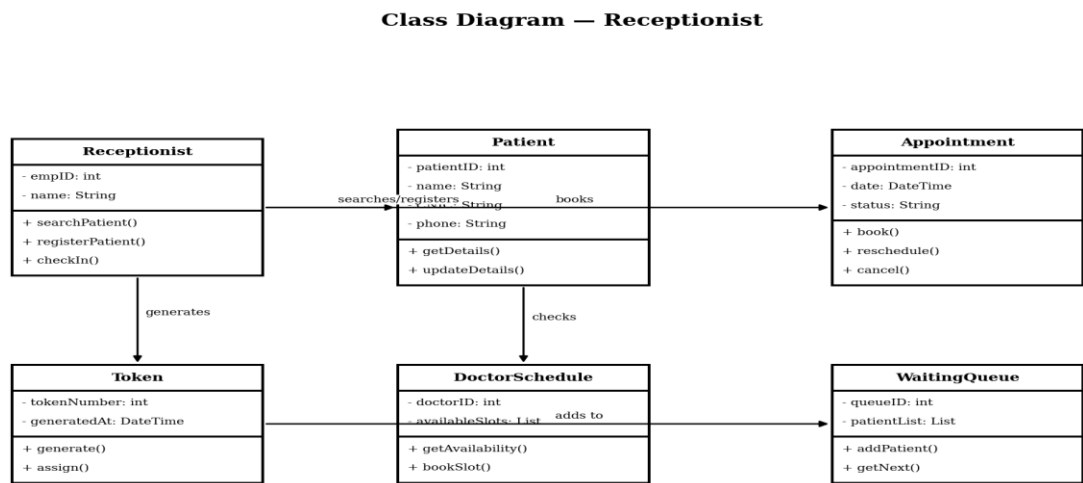
2.3 Sequence Diagram

Message flow between Receptionist, GUI, Database, and Appointment System for search, registration, booking, and check-in.



2.4 Class Diagram

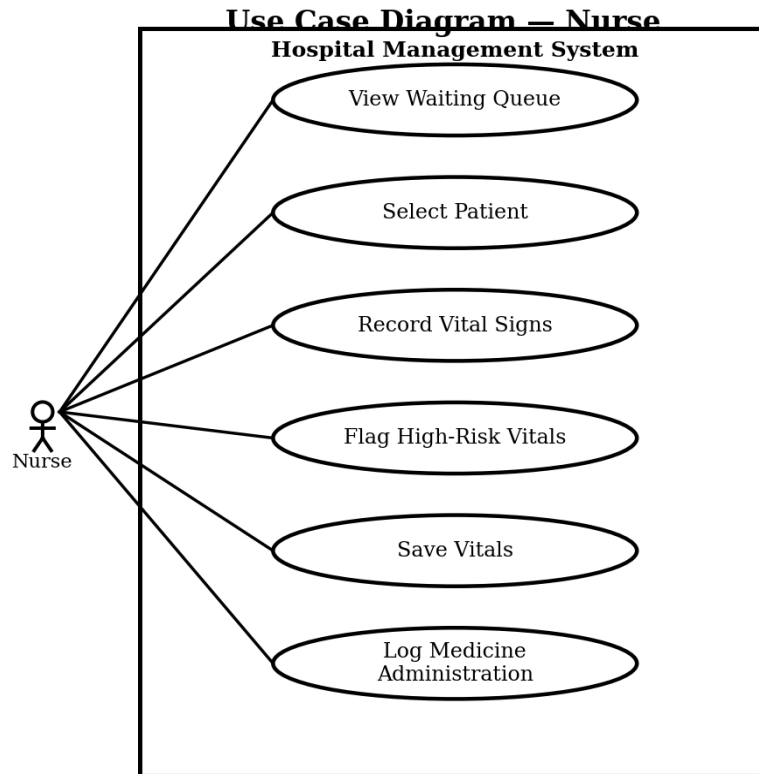
Receptionist, Patient, Appointment, Token, DoctorSchedule, and WaitingQueue classes with attributes and methods.



## Feature 3: Nurse (Triage & Ward Management)

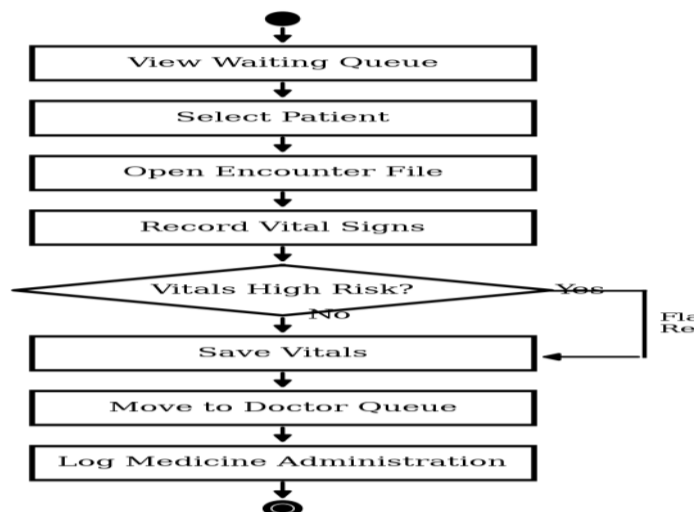
### 3.1 Use Case Diagram

Nurse can view waiting queue, select patients, record vital signs, flag high-risk vitals, save vitals, and log medicine administration.



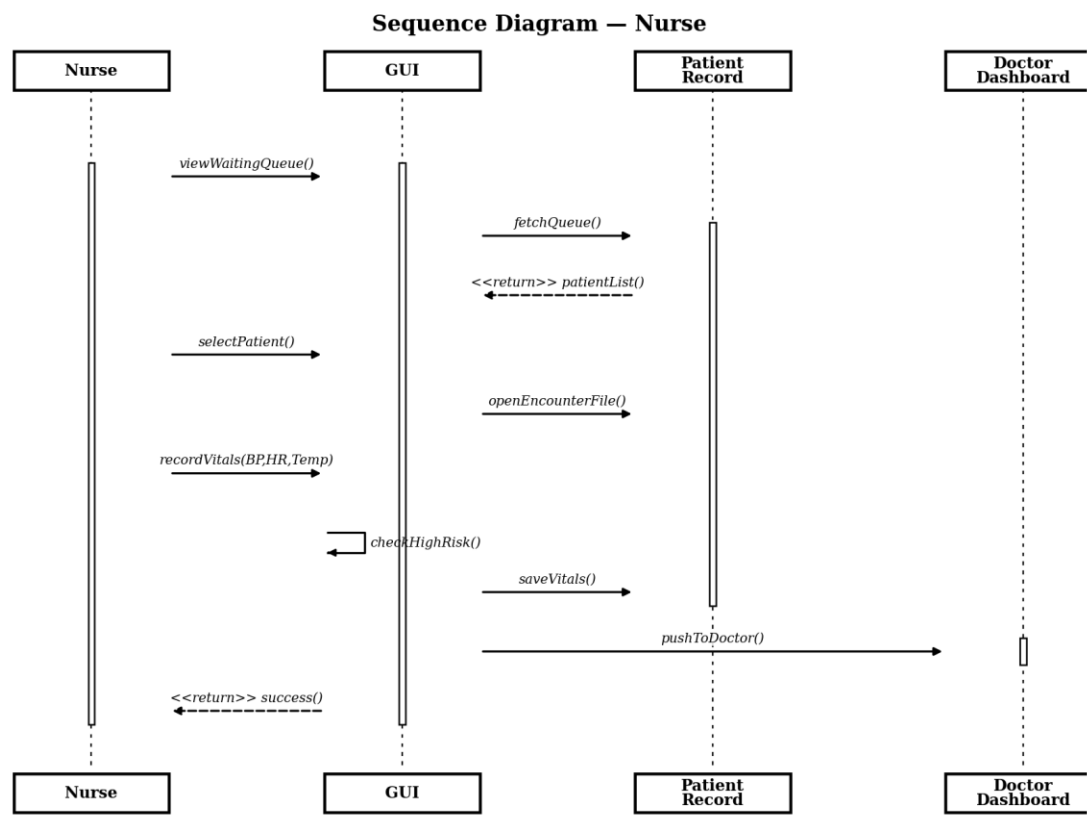
### 3.2 Activity Diagram

Nurse selects patient from queue, records vitals, flags high-risk vitals in red, saves vitals, moves patient to Doctor queue.



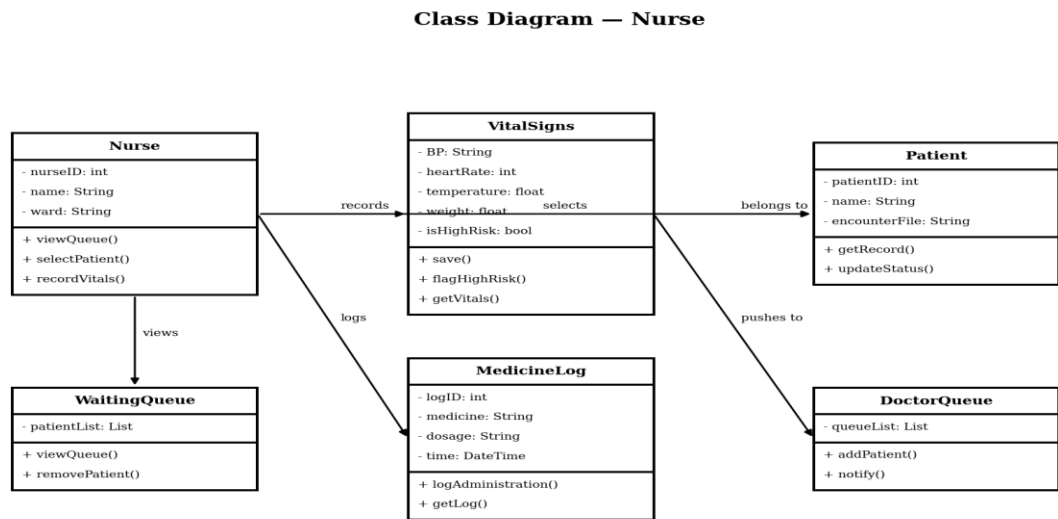
3.3 Sequence Diagram

Message flow between Nurse, GUI, Patient Record, and Doctor Dashboard for queue view, vital recording, and push to Doctor.



3.4 Class Diagram

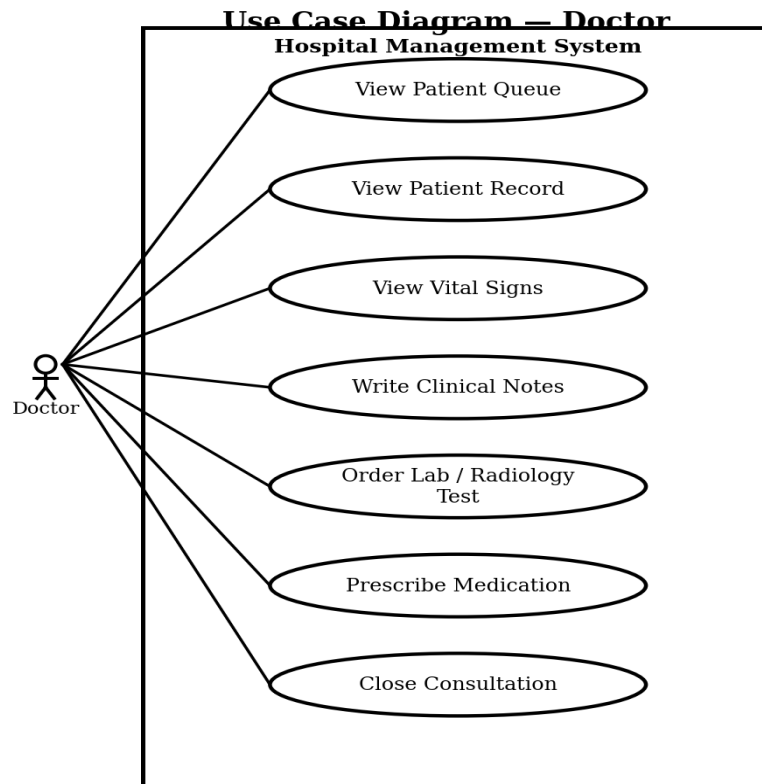
Nurse, VitalSigns, Patient, WaitingQueue, MedicineLog, and DoctorQueue classes with attributes and methods.



## Feature 4: Doctor (Clinical Decision Maker)

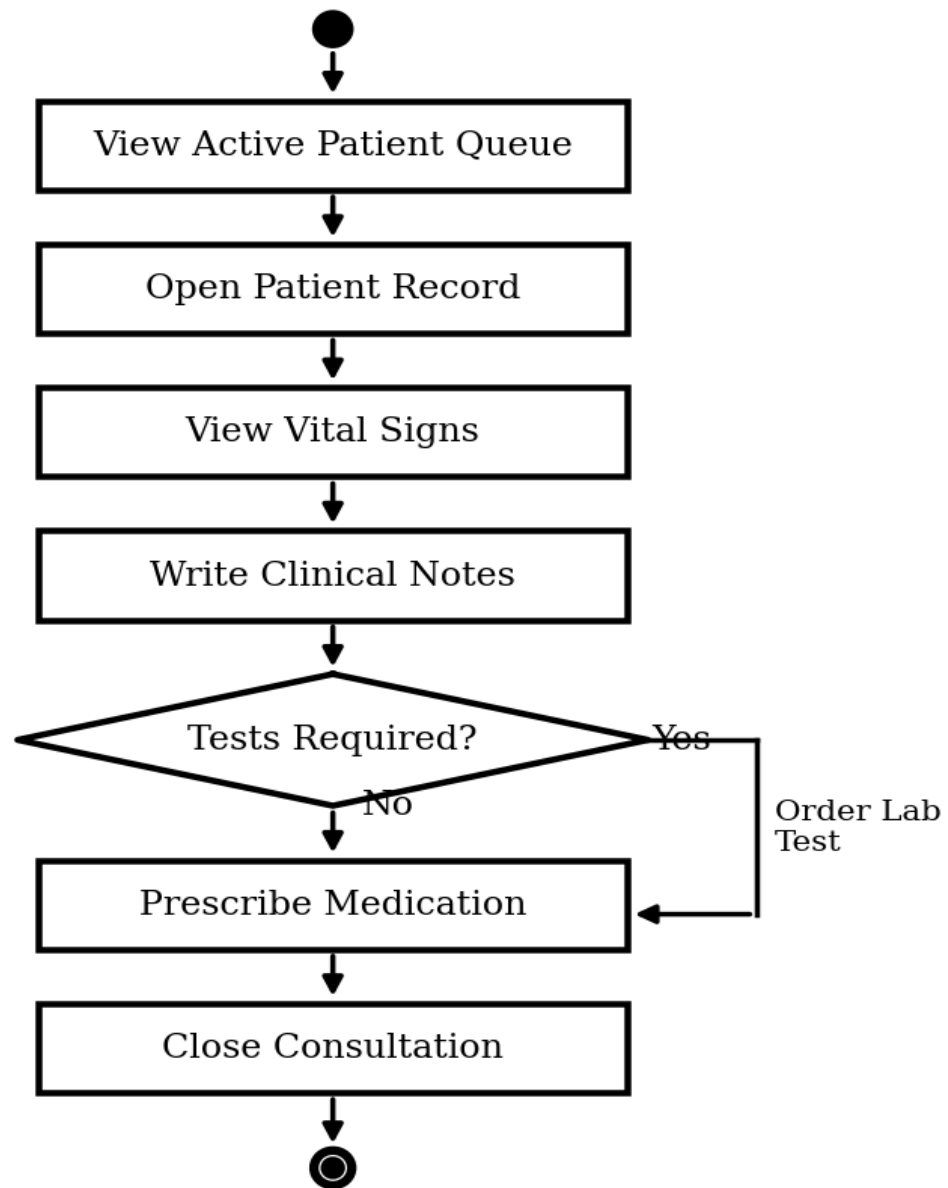
### 4.1 Use Case Diagram

Doctor can view patient queue, open records, view vitals, write clinical notes, order tests, prescribe medication, and close consultation.



### 4.2 Activity Diagram

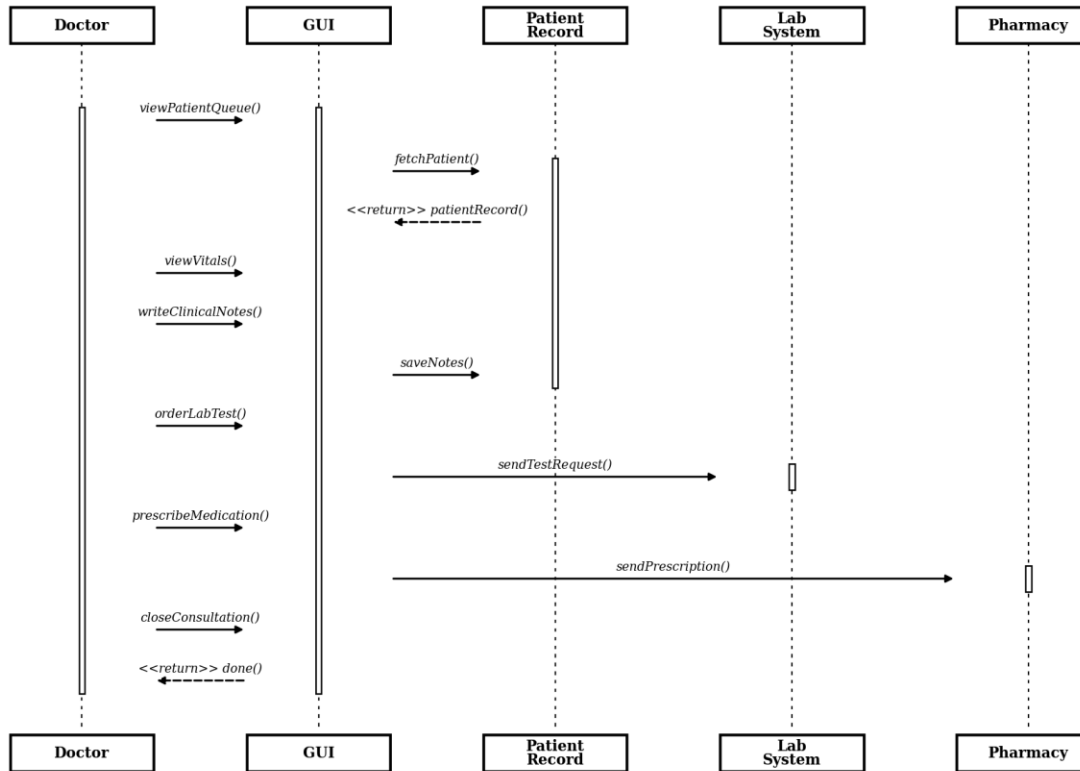
Doctor opens queue → views record → writes notes → orders tests if needed → prescribes medication → closes consultation.



#### 4.3 Sequence Diagram

Message flow between Doctor, GUI, Patient Record, Lab System, and Pharmacy for full consultation flow.

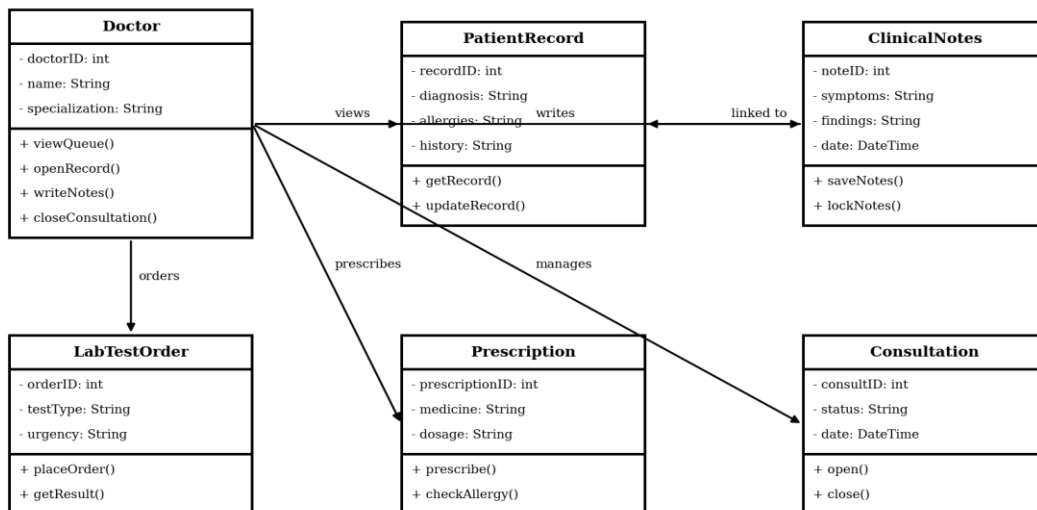
Sequence Diagram – Doctor



## 4.4 Class Diagram

Doctor, PatientRecord, ClinicalNotes, LabTestOrder, Prescription, and Consultation classes with attributes and methods.

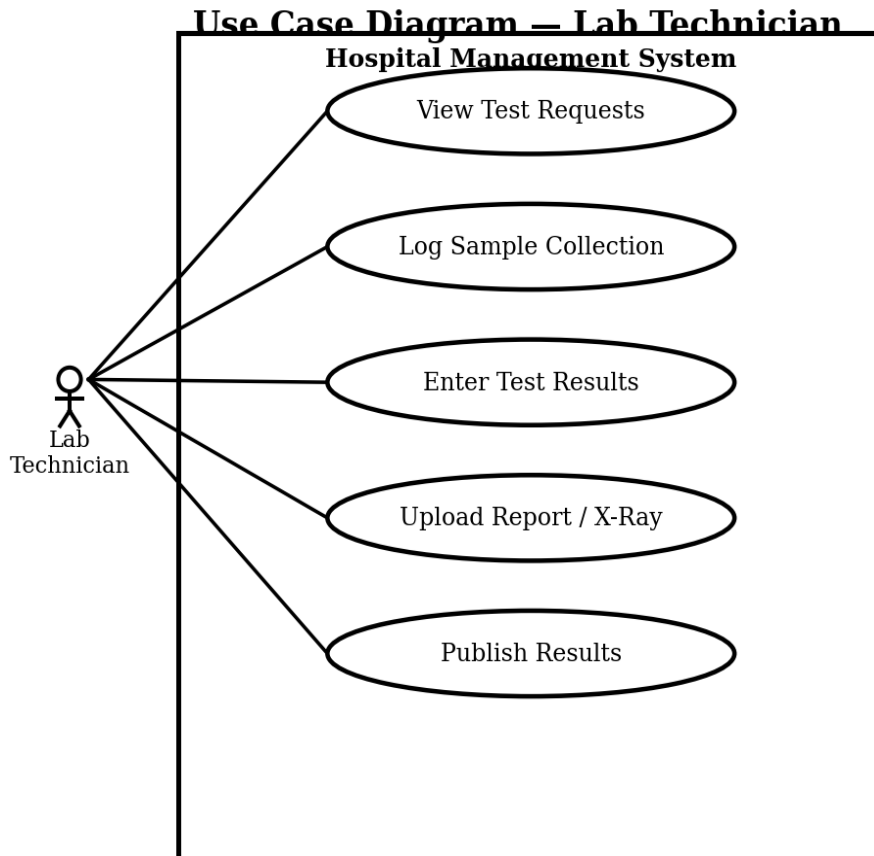
Class Diagram – Doctor



## Feature 5: Lab Technician / Radiologist (Diagnostics)

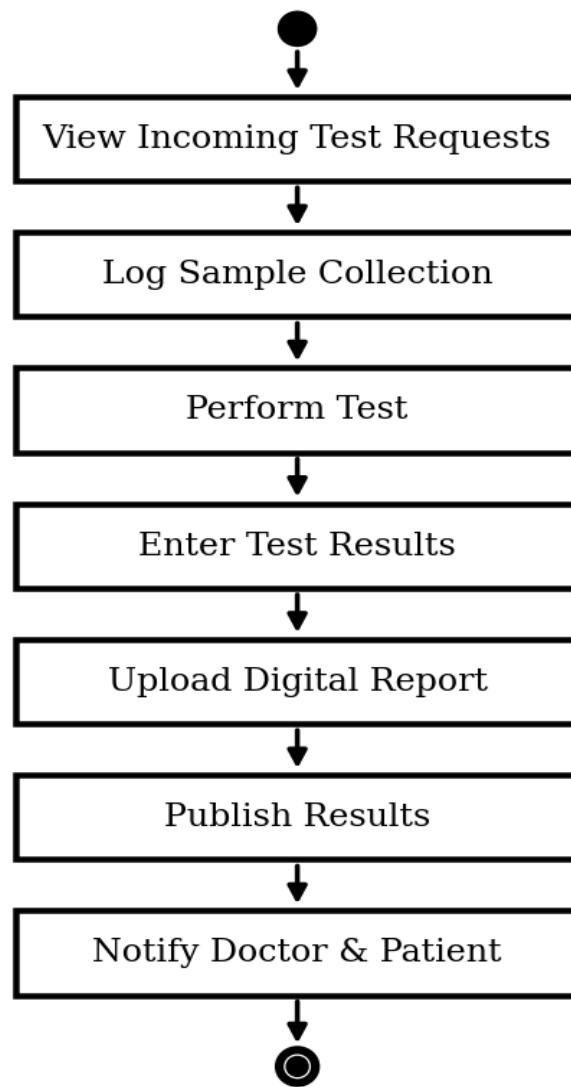
### 5.1 Use Case Diagram

Lab Technician can view test requests, log sample collection, enter results, upload digital reports, and publish results.



### 5.2 Activity Diagram

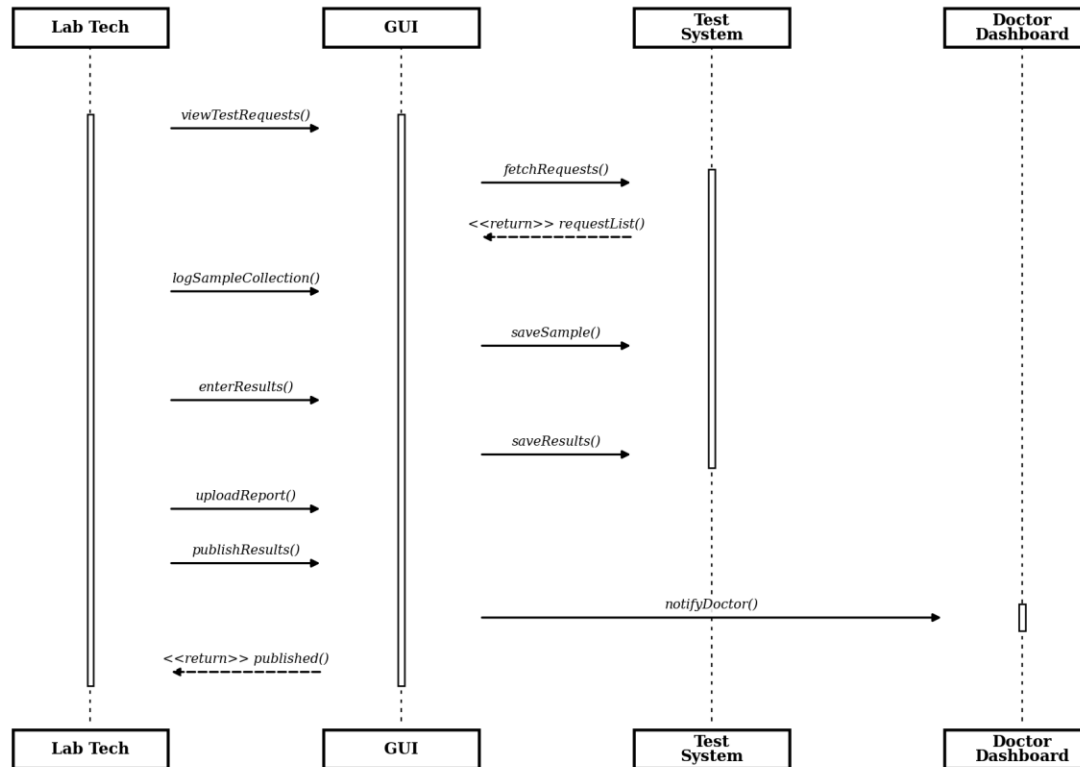
View requests → log sample → perform test → enter results → upload report → publish → notify Doctor and Patient.



### 5.3 Sequence Diagram

Message flow between Lab Tech, GUI, Test System, and Doctor Dashboard for request handling and result publishing.

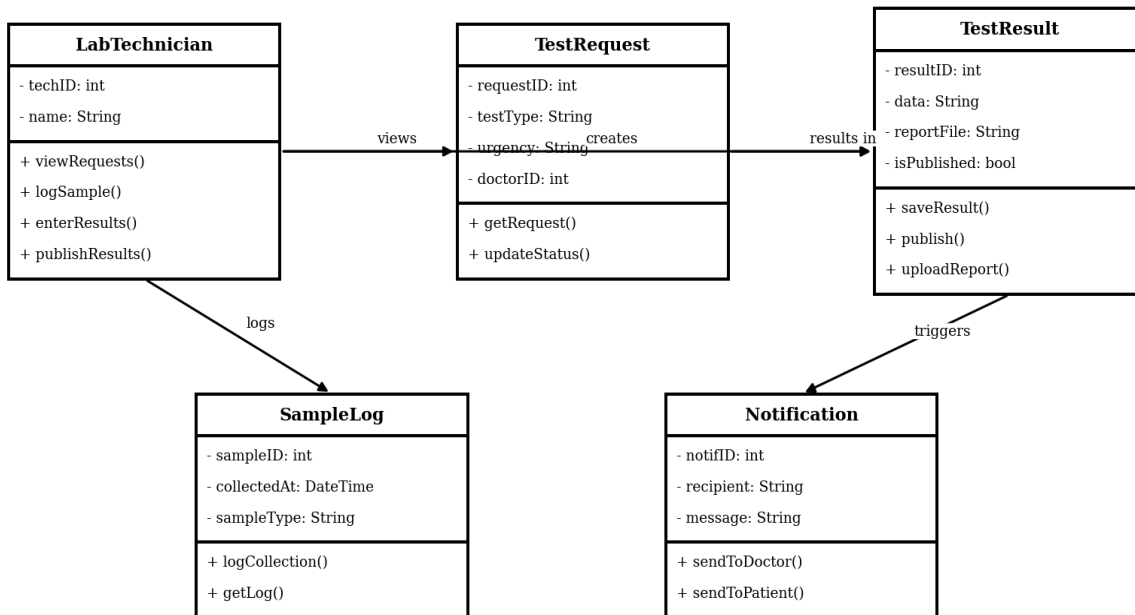
**Sequence Diagram — Lab Technician**



## 5.4 Class Diagram

LabTechnician, TestRequest, TestResult, SampleLog, and Notification classes with attributes and methods.

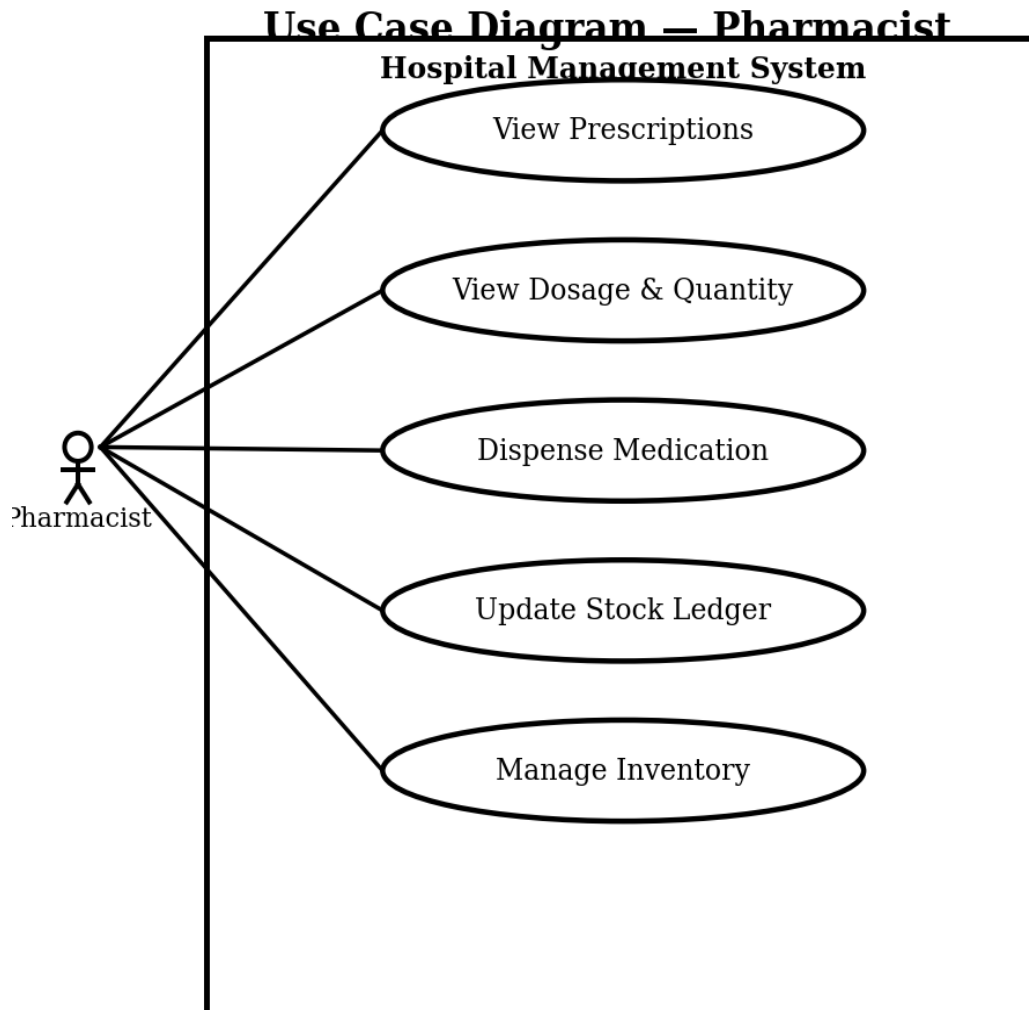
**Class Diagram — Lab Technician**



## Feature 6: Pharmacist (Dispensing & Stock Control)

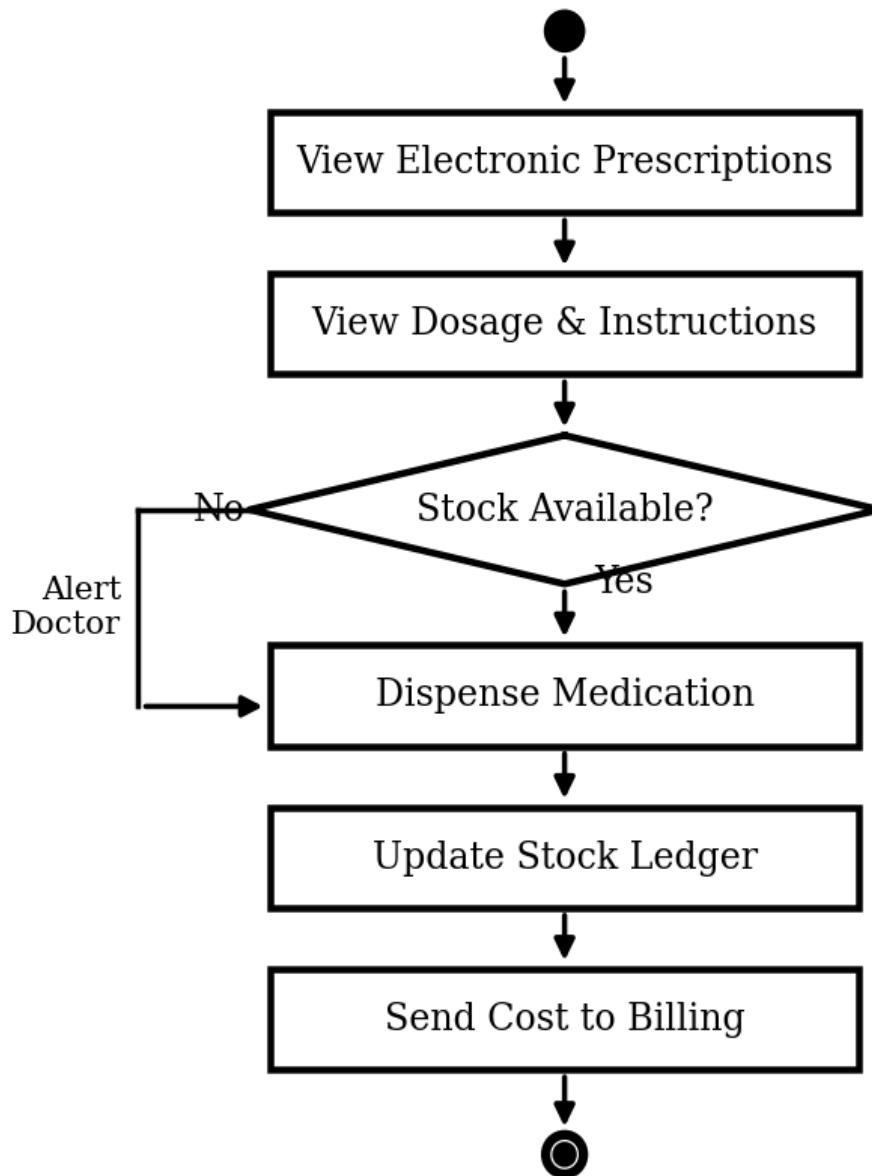
### 6.1 Use Case Diagram

Pharmacist can view prescriptions, check dosage and quantity, dispense medication, update stock ledger, and manage inventory.



## 6.2 Activity Diagram

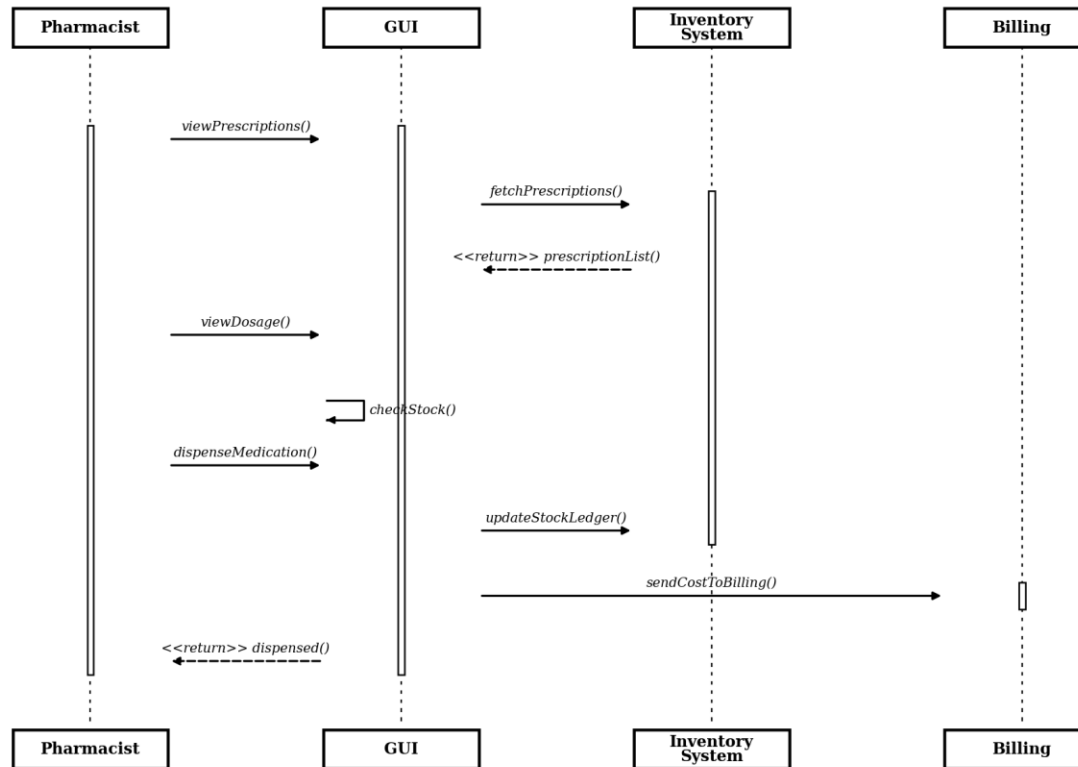
View prescription → check stock → dispense if available (alert Doctor if not) → update ledger → send cost to Billing.



## 6.3 Sequence Diagram

Message flow between Pharmacist, GUI, Inventory System, and Billing for prescription handling and dispensing.

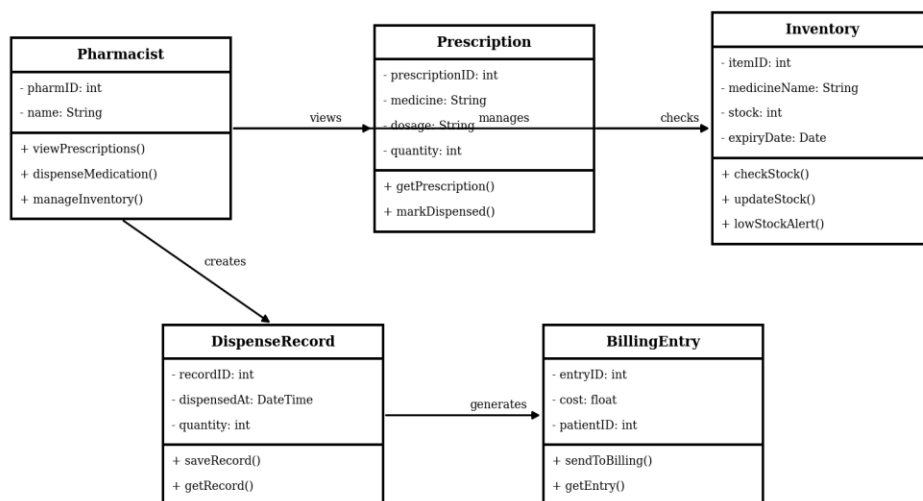
### Sequence Diagram – Pharmacist



## 6.4 Class Diagram

Pharmacist, Prescription, Inventory, DispenseRecord, and BillingEntry classes with attributes and methods.

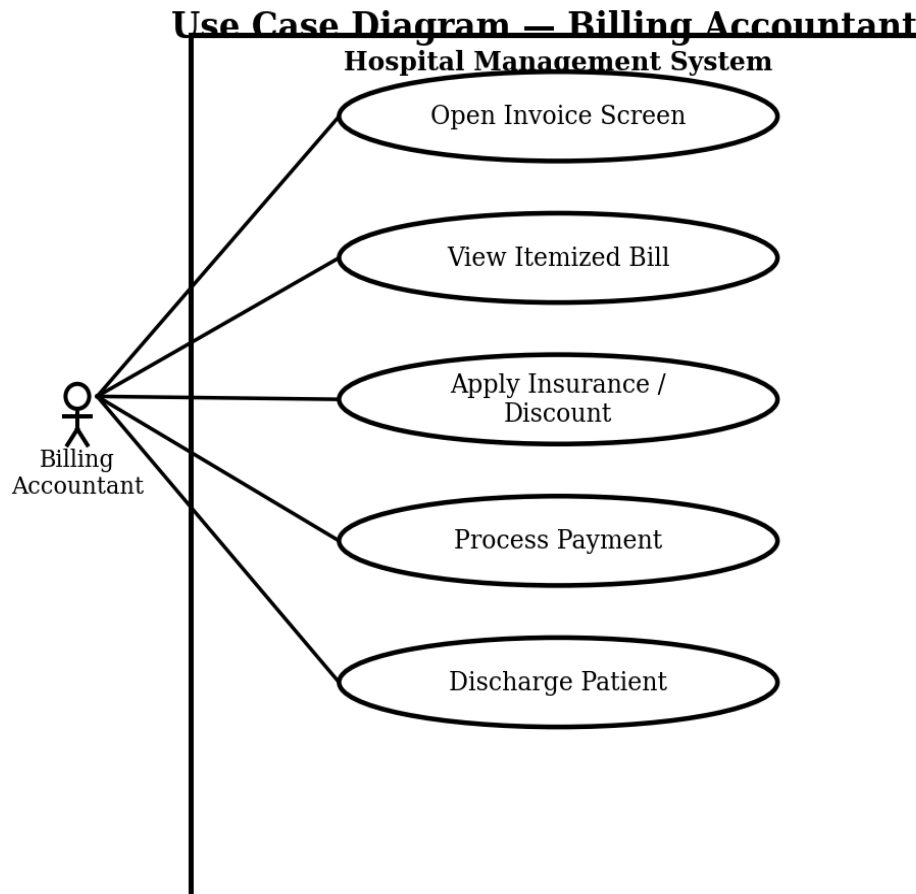
### Class Diagram – Pharmacist



## Feature 7: Billing Accountant (Finance & Checkout)

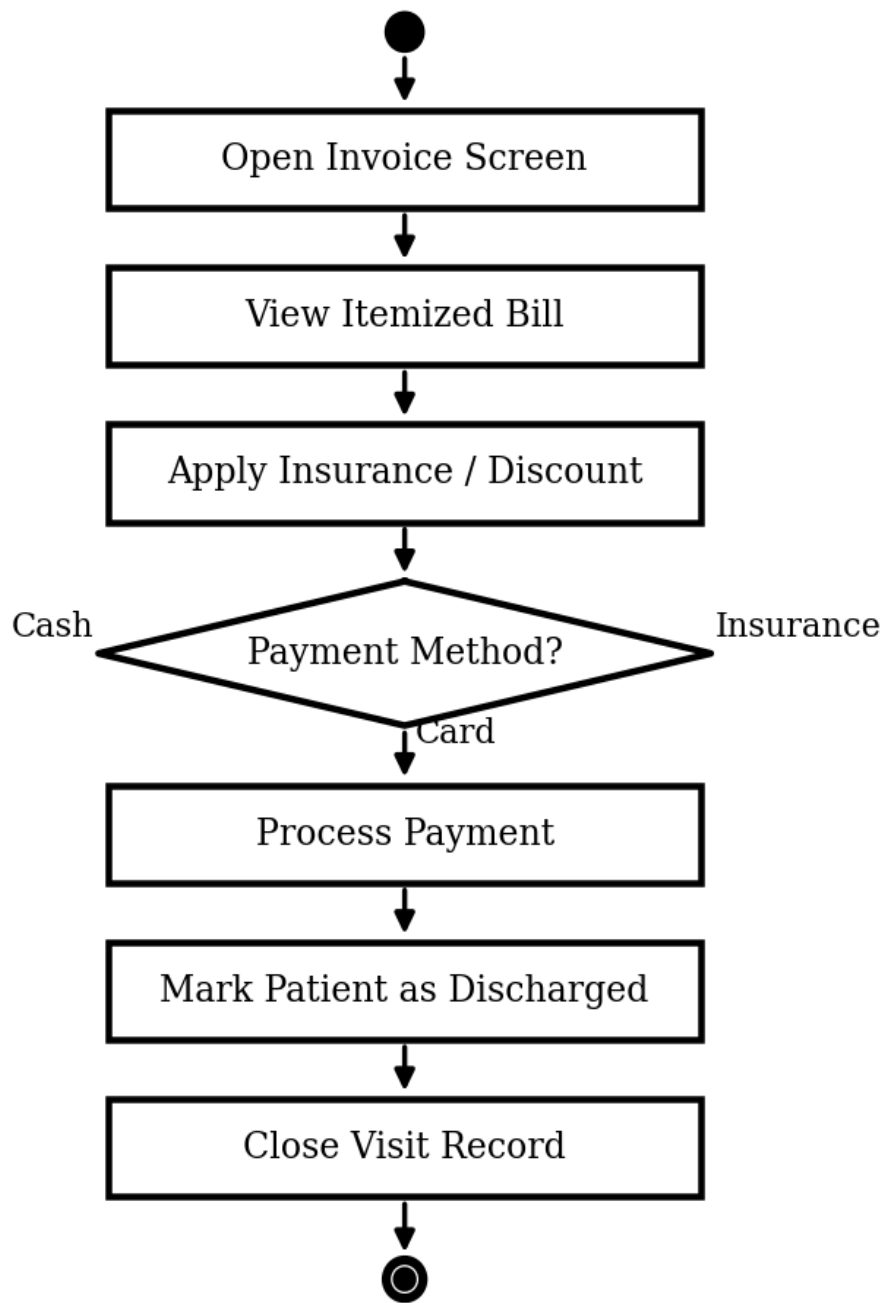
### 7.1 Use Case Diagram

Billing Accountant can open invoice, view itemized bill, apply insurance or discount, process payment, and discharge patient.



### 7.2 Activity Diagram

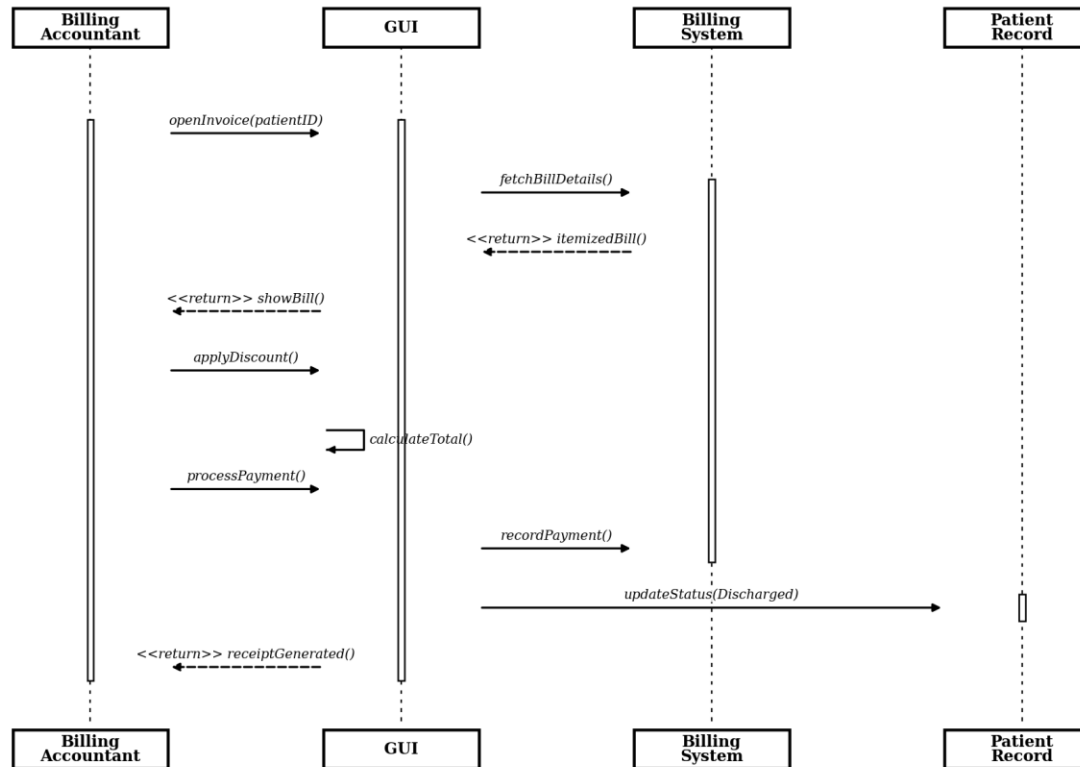
Open invoice → view itemized bill → apply discount → select payment method → process payment → discharge patient → close record.



### **7.3 Sequence Diagram**

Message flow between Billing Accountant, GUI, Billing System, and Patient Record for invoice, payment, and discharge.

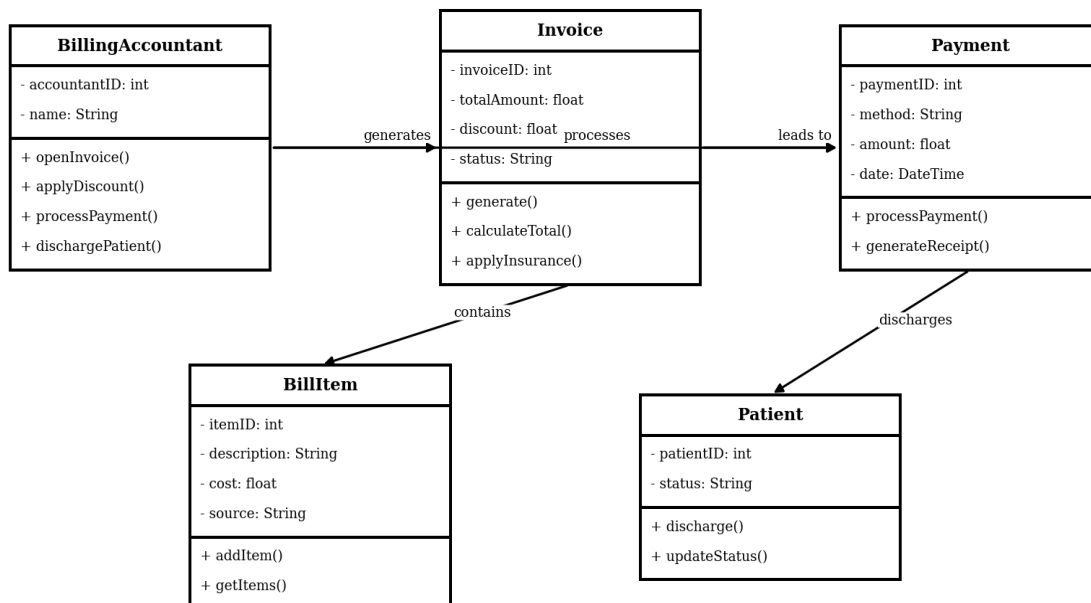
**Sequence Diagram – Billing Accountant**



## 7.4 Class Diagram

BillingAccountant, Invoice, Payment, BillItem, and Patient classes with attributes and methods.

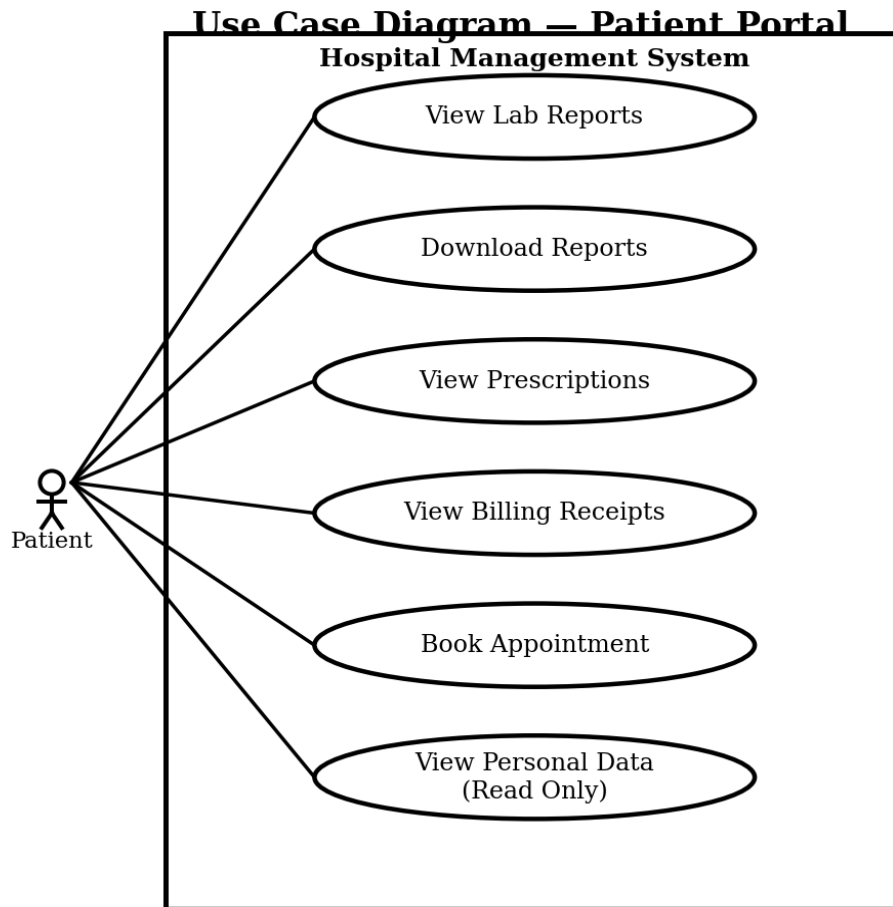
**Class Diagram – Billing Accountant**



## Feature 8: Patient Portal

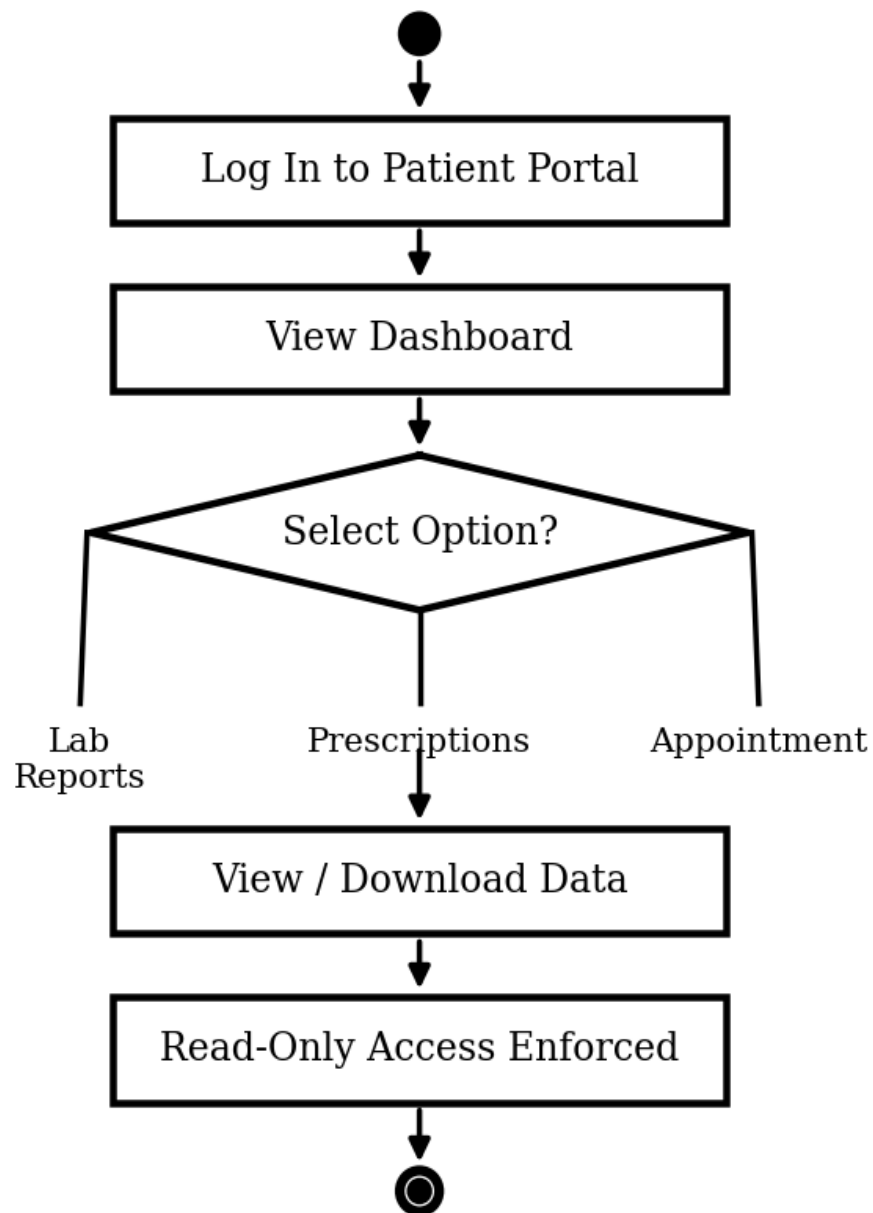
### 8.1 Use Case Diagram

Patient has read-only access. Can view lab reports, prescriptions, billing receipts, and book appointments online.



### 8.2 Activity Diagram

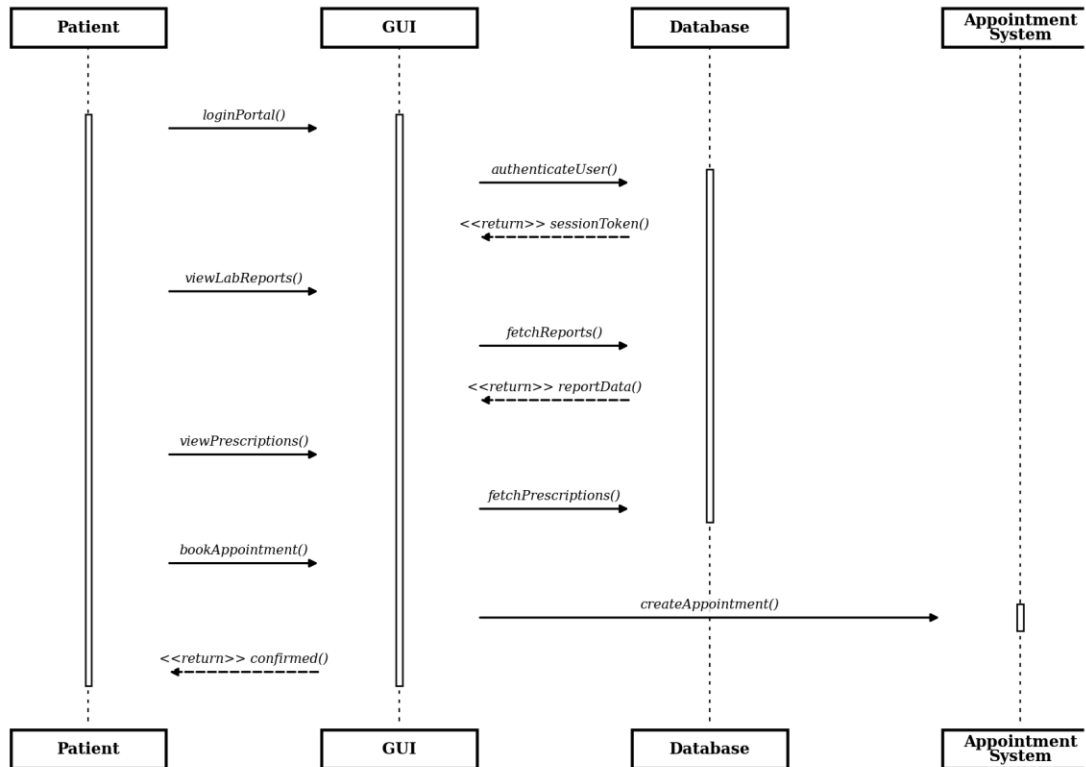
Patient logs in → views dashboard → selects option (Reports / Prescriptions / Appointment) → read-only access enforced.



### **8.3 Sequence Diagram**

Message flow between Patient, GUI, Database, and Appointment System for portal login and data access.

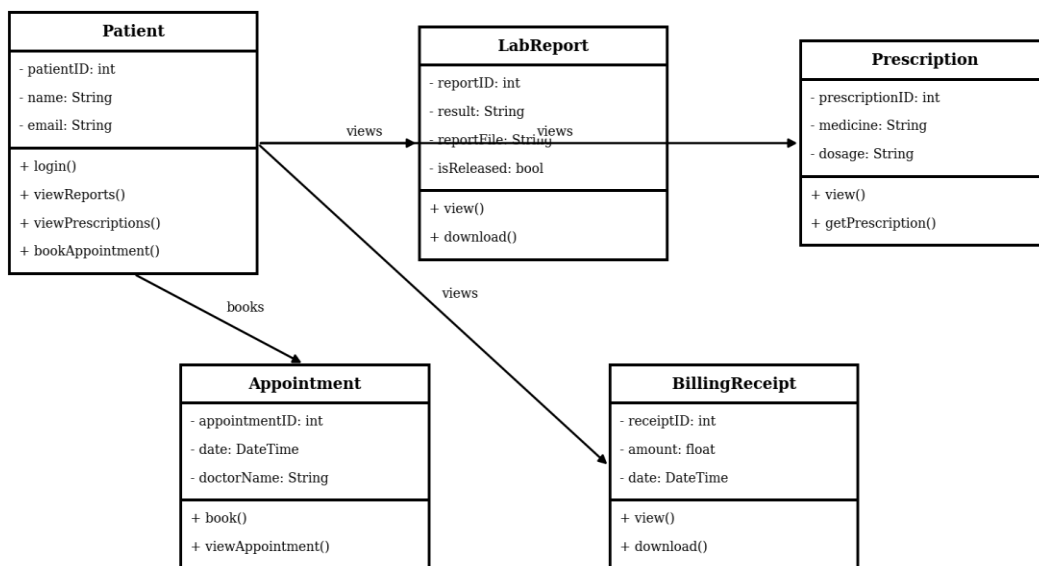
**Sequence Diagram – Patient Portal**



## 8.4 Class Diagram

Patient, LabReport, Prescription, Appointment, and BillingReceipt classes with attributes and methods.

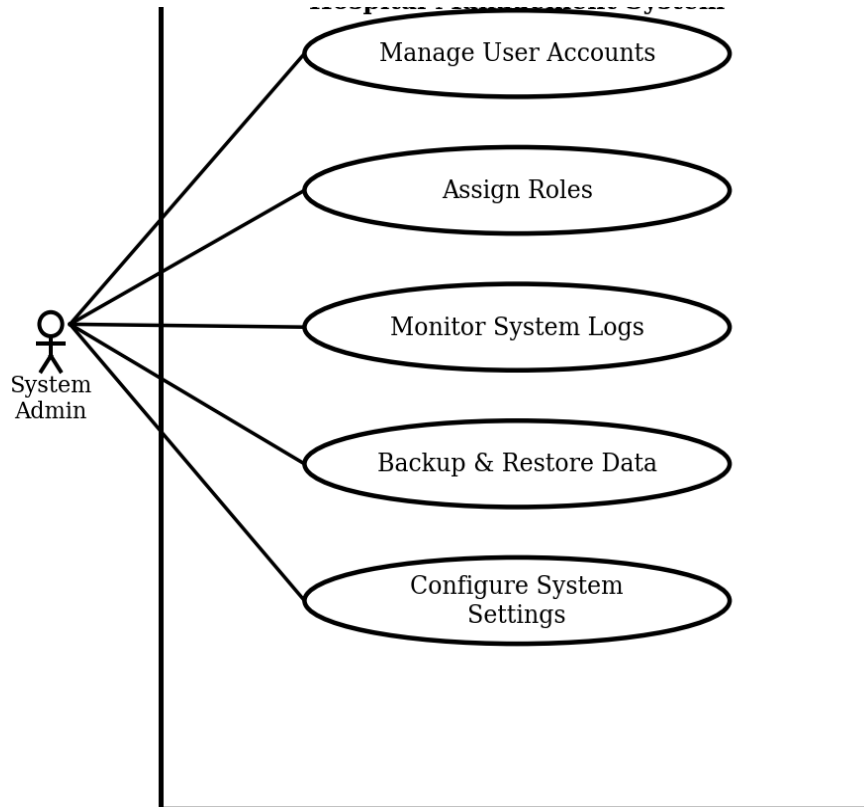
**Class Diagram – Patient Portal**



## Feature 9: System Administrator (IT & Security)

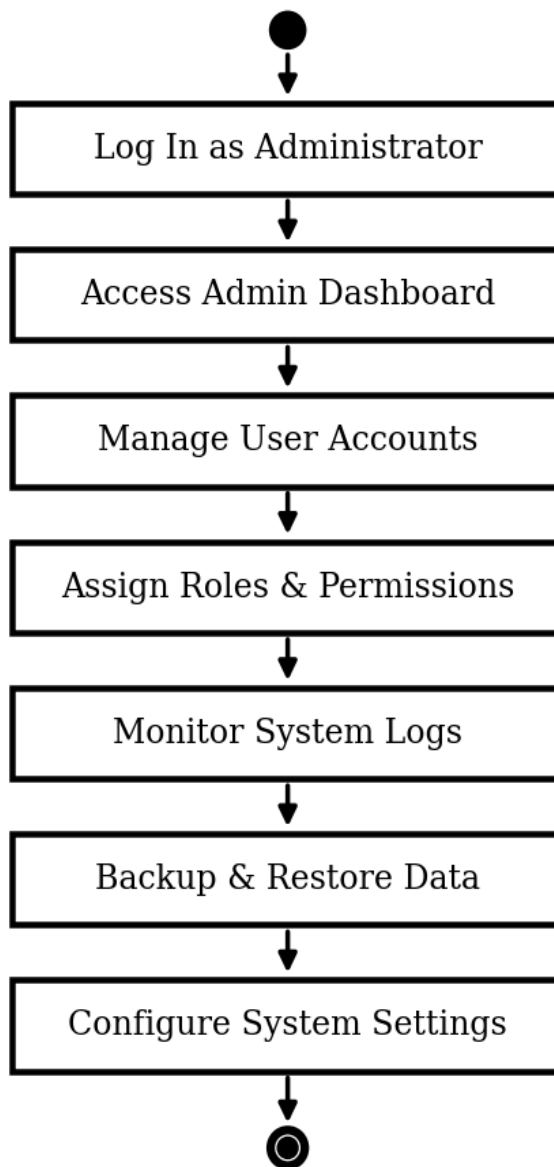
### 9.1 Use Case Diagram

Admin manages user accounts, assigns roles, monitors system logs, performs backups, and configures system settings.



### 9.2 Activity Diagram

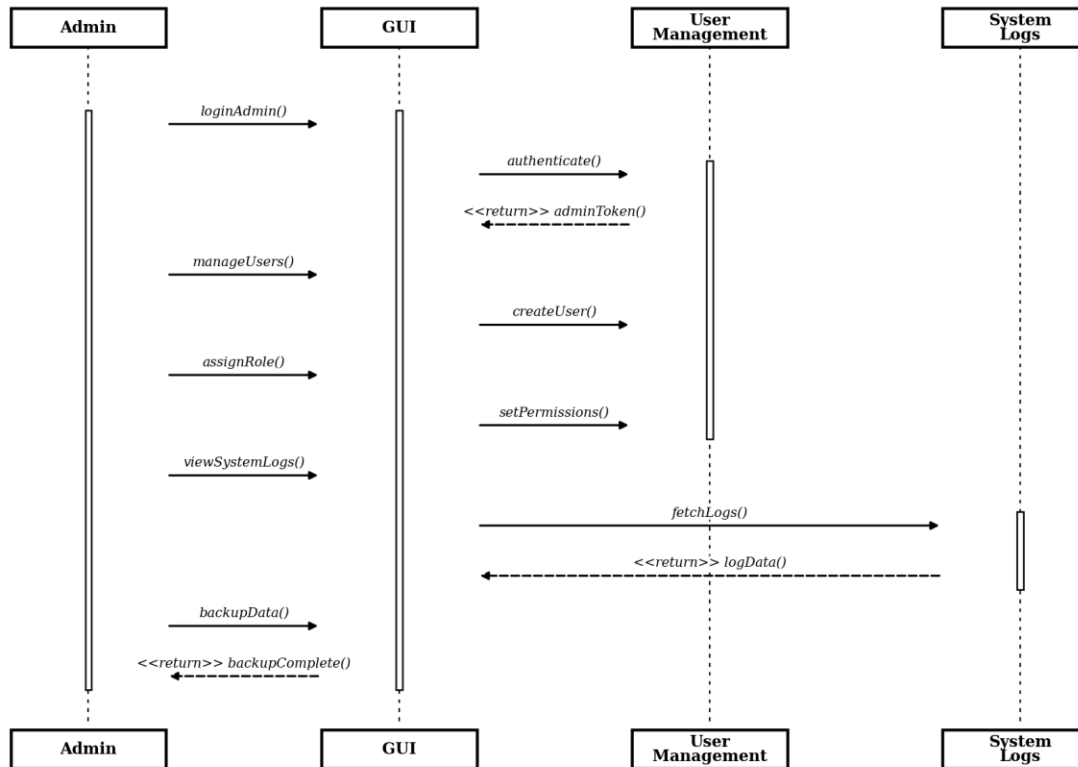
Admin logs in → accesses dashboard → manages users → assigns roles → monitors logs → backs up data → configures .



### **9.3 Sequence Diagram**

Message flow between Admin, GUI, User Management, and System Logs for user creation, role assignment, log monitoring.

**Sequence Diagram – System Administrator**



## 9.4 Class Diagram

SystemAdmin, UserAccount, Role, SystemLog, and BackupRecord classes with attributes and methods.

**Class Diagram – System Administrator**

